SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I K. Akshitha (192312527) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Akshitha

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1) Search-and-Rescue Drone

(i) why is online search reauired?

The drone does not know the complete building map. It discovers obstacles and safe paths while moving.

so, it must:

- observe the environment
- make a decision
- move
- Learn new information.
- Change its path when needed.

∴ Online search is suitable because the environment is unknown.

(ii) challanges of partial observability & dynamic environment.

Partial observability:

- The drone can see only a limited area.
- Some survivors may be hidden.
- Some obstacles may not be visible.

Dynamic environment:

- Debris may fall.
- Gas leaks may spread.
- New obstacles may appear.

Therefore, the drone must continuously update its decisions.

(iii) How does the drone update its internal model?

The drone uses sensors and cameras to collect information.

Steps:

1. Detect nearby objects and hazards.
2. Identify safe & unsafe areas.
3. Store this information in its map.

Thus, the drone gradually builds a map while exploring.

(iv) online Search vs uniformed & informed Search.

Search	meaning
uniformed Search.	Searches without extra knowledge.
informed Search.	Uses heuristic information to find a better path.
online Search.	Searches & learns while moving.

Examples:-

- BFS and DFS → uniformed Search.
- A* → informed search.

(v) Best approach

online informed Search is the best approach.

- The building is unknown
- The drone can learn while moving.
- It can avoid new hazards.

Conclusion

online Search is best because it provides adaptability, safety, and decision-making under uncertainly.

② Wedding Seating Arrangement - CSP.

(i) Variables, domains and Constraints.

Variables: Each guest is a variable.

Ex:- A, B, C, D → Guests.

Domains: The possible tables for each guest.

Ex:- Domain = {Table 1, Table 2, Table 3}.

Constraints:

1. Guests with conflicts cannot sit together.
2. Each table has limited Seats.
3. Family members should sit together.

The final Seating must satisfy all these constraints.

(ii) How does backtracking work?

Backtracking assigns guests one by one.

Steps:

1. Select a guest , 2. Assign a table , 3. check the constraints.
4. If correct, assign the next guest , 5. Try another table.

(iii) How does forward checking help?

Forward checking checks future possibilities early.

Ex:- If A is placed at Table 1 and B cannot sit with A, then Table 1 is removed from B's choices.

(iv) Challenges & best strategy.

challenges:

- Large number of guests.
- many tables.
- many constraints.

Backtracking + forward checking + Heuristics.

Conclusion:

CSP with constraint propogation is a good method for creating a valid seating arrangement efficiently.

3 Auction - Bidding AI - Minimax.

(i) How does minimax model the problem?

minimax is used when two players compete.

Here
- max = our AI
- min = opponent.

The AI tries to maximize its profit, while the opponent tries to reduce our profit.

The AI checks possible bids & opponent responses before choosing a bid.

(ii) Problems with large decision trees.

There can be many possible bids & ~~man~~ bidding rounds.

- The decision tree becomes very large.
- Searching takes more time.
- more memory is required.

(iii) How does Alpha-Beta Pruning help?

Alpha-Beta pruning removes branches that are not useful for the final decision.

- Reduces the number of nodes checked.
- Allows deeper searching.
- Gives the same minimax result.

(iv) Evaluation function.

The AI can calculate a score using:

Evaluation = profit + winning chance - Risk - Bid cost.

- Expected profit
- Chance of winning.
- Amount of money available.
- Bid cost.

(v) Best strategy.

The best approach is:

minimax + Alpha-Beta pruning + Evaluation Function.

The AI should search only a limited depth because of the time limit.

It can also use iterative deepening, where it searches:

Depth 1 → Depth 2 → Depth 3 → .

until the time limit is reached.

Conclusion

This approach gives a good balance b/w good decisions, less computation, and fast response.